

Tools & remarks

Convenient tools

- Generic tools
 - [Napari](#): interactive multidimensional image viewer and annotation tool, with a rich plugin ecosystem
- Segmentation tools (ML based)
 - [StarDist](#): robust nucleus and cell instance segmentation using star-convex shapes
 - [Cellpose](#): generalist deep-learning segmentation for many cell types
 - [micro-sam](#): SAM-based interactive and automated segmentation workflows for microscopy
 - [OrganoidTracker](#): tracking and lineage analysis for organoid time-lapse data
 - [ilastik](#): pixel/object classification and segmentation with minimal coding
 - [biapy](#): (generic) deep learning-based segmentation and tracking for bioimage analysis

Additional tools (*suggested by LLM*):

- [SimpleITK](#): medical/microscopy image analysis tool package
- [Squidpy](#): spatial omics analysis with image feature extraction support

Convenient non-python tools:

- [Fiji/ImageJ](#): classic GUI platform with many plugins and macro support
 - [Labkit](#): interactive machine learning segmentation in Fiji
 - [TrackMate](#): particle/cell tracking in Fiji
- [CellProfiler](#): pipeline-based high-throughput image analysis and feature extraction

An extensive list of bio-image analysis tools:

- <https://github.com/hallvaaw/awesome-biological-image-analysis>